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DATA DESCRIPTOR

A Comprehensive Hyperspectral Image Dataset for Forest Fire Detection and Classification

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In this paper we introduce a new large-scale hyperspectral satellite image dataset named OHID-FF, specifically designed for forest fire detection and classification tasks. The OHID-FF dataset comprises 1,197 hyperspectral images from 22 different scenarios, with each image featuring 32 spectral bands and a spatial resolution of 10 meters per pixel. The dataset covers 22 locations in Australia, encompassing urban areas, mountainous regions, oceans, and other terrains. Compared to existing fire datasets, OHID-FF offers a richer volume of data and higher imaging quality, making it an ideal choice for training deep neural networks. Through benchmark experiments on this dataset, we found that existing methods face challenges in accurately classifying OHID-FF data, setting a new benchmark for hyperspectral imaging classification. Additionally, we provide detailed descriptions of the dataset preparation process, data sources, tile creation, and annotation procedures. Furthermore, we present experimental results using different deep learning models for fire detection and image classification, demonstrating the potential of this dataset in practical applications.

Background & Summary

Forest fires constitute a grave danger to human society and the natural environment, characterized by a high degree of randomness in their occurrence, which makes predicting their exact timing and location extremely challenging. Since the introduction of satellite remote sensing technology in the 1970s, this advanced technology has swiftly been adopted for fire monitoring, leveraging its formidable macro-observation capabilities¹. Over the course of nearly half a century, it has accumulated substantial application experience and driven technological innovations. Satellite remote sensing operates by detecting the radiation intensity of thermal infrared electromagnetic waves emitted from the Earth's surface to monitor fire conditions, thereby capturing thermal anomalies at the macro level. However, remote sensing data alone cannot directly ascertain specific fire scenarios; they can merely identify the presence of thermal anomalies on the Earth's surface and estimate their intensity in a general sense. The types of thermal anomalies detected by remote sensing are varied, encompassing volcanic eruptions, forest and grassland fires, oil spills and natural gas combustion, agricultural straw burning, industrial thermal emissions, and others². Notably, while volcanic eruptions and oil spill combustion also produce thermal anomalies, they do not necessarily result in fires. Nevertheless, forest and grassland fires, along with agricultural straw burning, pose significant damage to human society and the natural environment, and their occurrence probability and frequency are heightened in specific regions and seasons. Consequently, they have emerged as primary areas of concern in current satellite remote sensing fire monitoring efforts³.

Traditional fire detection technologies are refined and implemented around three categories^{4,5}. The first category is identifying high-temperature pixels through threshold-setting⁶. The second category is adopting a contextual comparison approach^{7,8} to contrast detected high-temperature pixels against their surrounding backgrounds. And the last category is detecting fire pixels based on smoke emissions and flame dynamics. The threshold method often fails due to variations in conditions, while the contextual comparison method faces challenges from temperature fluctuations. Processing low-resolution RS datasets exacerbates misidentification

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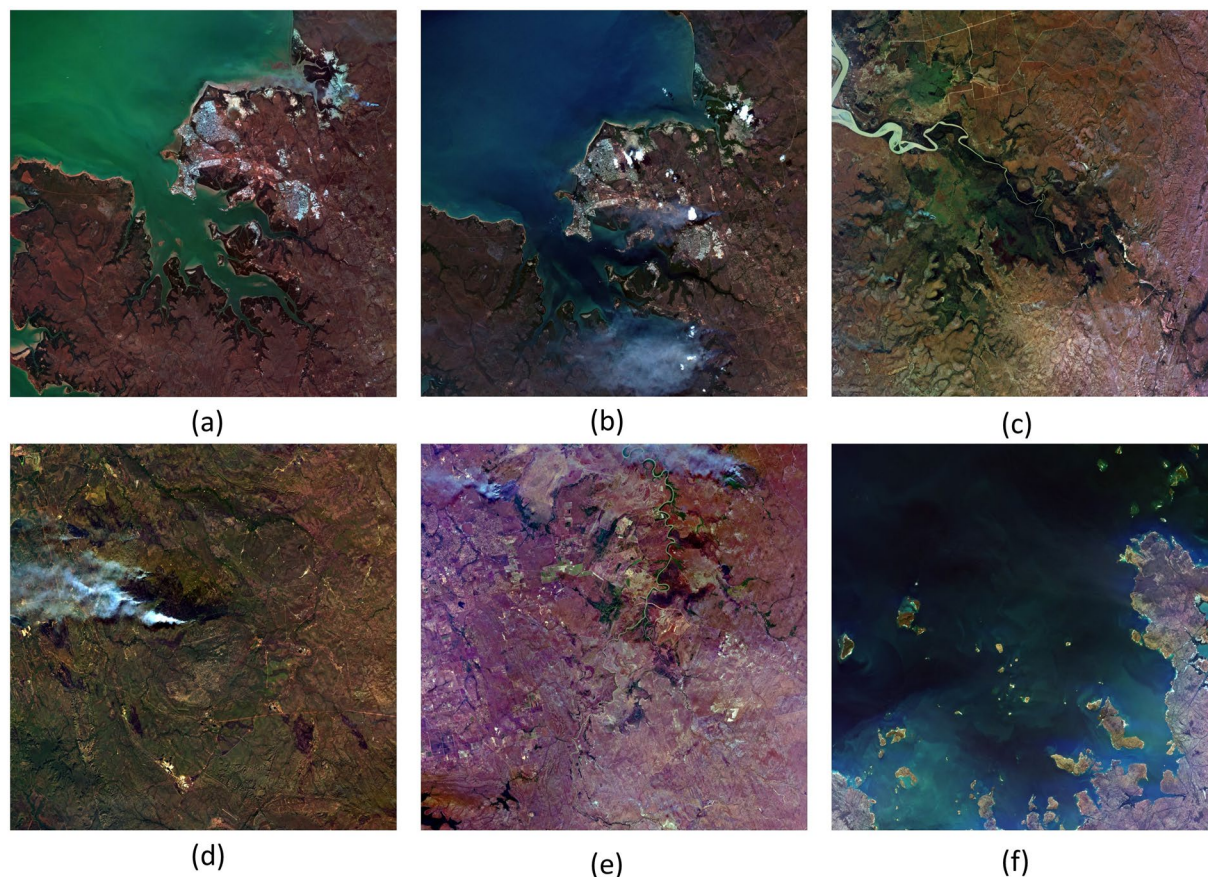


Fig. 1 Original remote sensing imagery from the OHID-FF dataset: **(a,b)** Remote sensing imagery of coastal areas captured at two different time points. **(c–e)** Typical forest fire regions, with additional geographical features included in the images, such as cities and rivers. **(f)** Background marine area without any fire points, where smoke may easily lead to confusion.

issues⁴. Those methods depend on conventional Machine Learning (ML) methods⁹ which necessitate detailed and lengthy feature extraction processes.

In recent years, deep learning has demonstrated powerful capabilities in image classification¹⁰ and object detection^{11–13}. The task of remote sensing image classification can determine whether the image depicts a fire scene, while object detection can pinpoint the exact location of the fire within the image. The key advantage of deep learning-based methods over traditional ones lies in their ability to automatically learn data features at more abstract levels through deep architectures, without the need to design specific manual features for particular image data or classification methods. This significantly enhances the effectiveness of image classification.

Image classification networks typically utilize state-of-the-art (SOTA) backbone networks to extract image features, and then employ these deep features to accomplish image classification tasks, such as using ResNet¹⁴ or ViT¹⁵. Common object detection methods based on deep learning can be divided into two-stage and one-stage approaches¹⁶. Two-stage methods divide the object detection task into two phases: the first phase typically generates candidate object regions, while the second phase classifies and verifies these candidate regions. Although two-stage methods improve accuracy and robustness through the cooperation of the two phases, they generally require more computational resources and time. On the other hand, one-stage methods directly detect objects from the input image, making them generally simpler and more efficient, but they may compromise detection accuracy and recall rates in complex scenes. Common two-stage object detection algorithms mainly include the RCNN series¹⁷, while one-stage object detection methods primarily encompass the YOLO series^{18,19}.

Dataset Overview

In this paper, we introduce OHID-FF²⁰, a new large-scale remote sensing dataset for forest fire detection and classification. This dataset comprises 22 original remote sensing images collected from diverse scenarios, each containing 32 spectral bands and characterized by a high spatial resolution of 10 meters, as shown in Fig. 1. The satellite-acquired images are large, with each scene measuring 5056×5056 pixels and reaching gigabyte-scale file sizes. Direct processing of such large volumes of data places substantial demands on memory, storage, and computational resources. To mitigate these challenges, the original images are partitioned into smaller 512×512 pixel patches through a tiling procedure, forming the basis of the subsequent dataset for efficient processing and analysis. With its abundant data volume and high-quality imaging characteristics, the OHID-FF dataset²⁰ presents new opportunities for remote sensing monitoring of forest fires. Figure 1 showcases an example of an

Dataset	Resolution	Channels	Spectrum	Number of images	Number of scenes	Size (pixels)	Data source
FASDD_RS ²¹	10 m	13	400–2500 nm	2223	10	1000 × 1000	Sentinel-2
CAF_SmokeSEG ²²	16 m	8	400 nm–890 nm	240	93	512 × 512	GF-6
Landsat-8 dataset ²³	RGB (30 m) SWIR (30 m) NIR (30 m) TIRS (60 m) PAN (15 m)	11	400–1390 nm	47	47	*	Landsat-8
Wildfire dataset ²⁴	*	*	*	2700	*	4057 × 3155 (average)	government databases, Flickr, and Unsplash
AIDER ²⁵	—	3	—	2545	—	240 × 240	world-wide-web, google images, bing images
FLAME ²⁶	—	3	—	2003	—	3480 × 2160	Phantom 3 Professional
FLAME2 ²⁷	—	4	—	53451	—	3480 × 2160 254 × 254	Mavic 2 Enterprise Advanced dual RGB/ IR camera
OHID-FF ²⁰	10 m	32	400–1000 nm	1197	22	512 × 512	CMOS (spaceborne)

Table 1. Comparison of different remote sensing fire point datasets. *Paper²³ on the Landsat-8 dataset don't specify image size. Paper²⁴ on the Wildfire dataset don't specify resolution, channels, spectrum and scenes.

original remote sensing image from the OHID-FF dataset²⁰. It is an open remote sensing fire dataset covering multiple scenarios and suitable for various purposes. Benchmark experiments for image classification and object detection conducted on the OHID-FF dataset²⁰ indicate that existing methods face challenges in accurately classifying its data. Consequently, OHID-FF²⁰ establishes a new benchmark for forest fire detection.

The FASDD_RS dataset²¹ stands out in fire detection due to its extensive data volume and high-resolution imagery, albeit with certain drawbacks, including data imbalance, annotation errors, limited geographical and seasonal coverage, lack of dynamic information, and substantial computational demands. Similarly, the CAF_SmokeSEG dataset²² demonstrates insufficient data diversity and poor representation of thin smoke pixels at edge regions. The Landsat-8 dataset²³ exhibits exceptional performance in detecting and analyzing forest fire smoke, offering multi-temporal and multi-regional coverage along with high-resolution imagery. However, it also faces challenges such as data imbalance, annotation inaccuracies, geographical and seasonal constraints, absence of dynamic information, and considerable computational requirements. The Wildfire dataset²⁴, compiled from public images, presents additional challenges: inconsistent resolutions, watermarks, and potential scene duplication (e.g., unlabeled multi-angle views of the same fire event). While thorough preprocessing is essential, the associated study lacks detailed documentation on data cleaning procedures. The AIDER dataset²⁵, with only 320 images of fire/smoke, suffers from an excessively small original dataset size, potentially relying on augmentation and thereby risking overfitting. The FLAME dataset²⁶ boasts significant advantages in multimodal data, high-resolution annotations, and real-scene coverage, yet it exhibits geographical homogeneity by being exclusively collected from pine forests in Arizona during winter (at 6 °C, windless), lacking diversity in climate conditions (such as strong winds, rain, and snow) and vegetation types, thus limiting its generalization capability. Additionally, its scene specificity, focusing solely on pile burns and neglecting dynamic, spreading natural wildfires (broadcast fires), necessitates future expansion. FLAME2 dataset²⁷ fills a gap in the field of drone-based fire monitoring with multimodal high-resolution data and rigorous annotations, although improvements are still needed in terms of geographical and scene coverage, annotation granularity, and openness²⁸. In comparison with other remote sensing fire datasets, as presented in Table 1, our OHID-FF dataset²⁰ distinguishes itself in terms of data size, scene diversity, and annotation accuracy, providing a more comprehensive and in-depth data foundation for fire detection research and demonstrating remarkable research value and potential.

Methods

The method for preparing the OHID-FF dataset²⁰ can be divided into three main steps, as illustrated in Fig. 2. In the first step, we completed the experimental setup and utilized the Zhuhai-1 satellite to collect hyperspectral data. The Zhuhai-1 satellite employs a linear array push-broom imaging method, capable of acquiring imagery with high spectral and spatial resolution. In the second step, we processed the collected large-sized original satellite imagery by tiling it into smaller segments. By dividing the large imagery into multiple smaller image patches, we not only improved the efficiency of data processing but also reduced the demand for computational resources. Finally, in the third step, we annotated the sliced imagery to construct a high-quality dataset for training and validation of deep learning models. This process included precise annotation of forest fire targets, ensuring that the dataset could effectively support hyperspectral image classification and forest fire detection tasks.

Data acquisition. The imagery in OHID-FF dataset²⁰ originates from the Zhuhai-1 satellite constellation, designed and produced by Zhuhai Orbita Aerospace Science and Technology Co., Ltd. In 2023 company has been renamed to Zhuhai Aerospace Microchips Science & Technology Co., Ltd. Zhuhai-1 satellite constellation comprises multiple satellites, including hyperspectral satellites, with several already launched. The relevant sensor information for the Zhuhai-1 satellite is presented in Table 2. These satellites employ a linear array push-broom imaging method, continuously scanning along the orbital direction with a linear array detector to achieve hyperspectral imaging of the Earth's surface. With a spectral resolution as high as 2.5 nanometers, they can differentiate subtle spectral characteristics of surface materials, and a spatial resolution of 10 meters enables clear identification of surface details. Their extensive coverage allows for global data acquisition. The satellites support multiple imaging modes, including standard, rapid, and hyperspectral imaging modes. The data from Zhuhai-1 hyperspectral

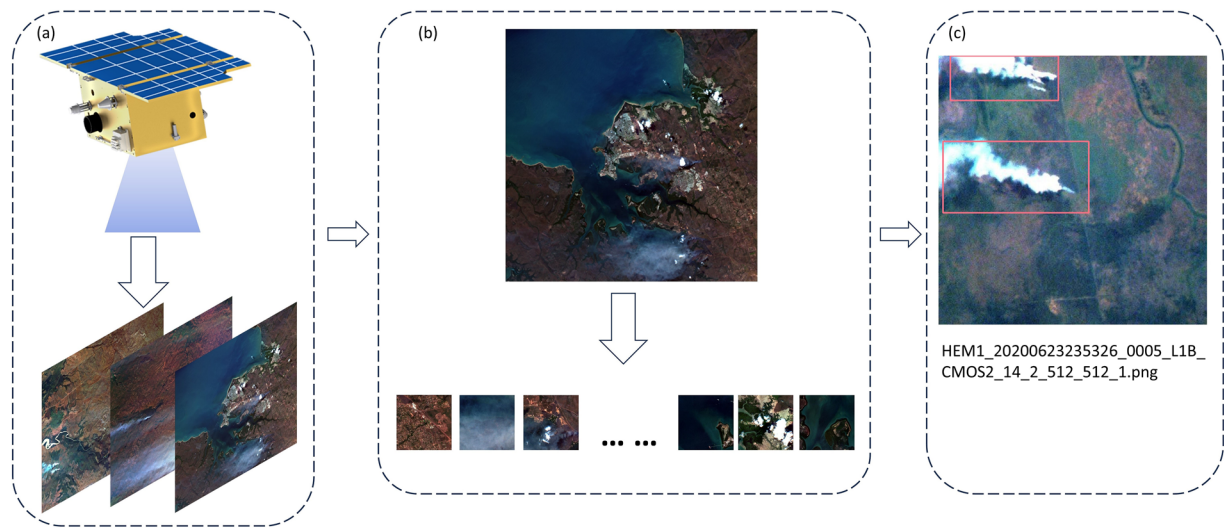


Fig. 2 Flowchart for the creation of OHID-FF dataset: **(a)** Schematic diagram of satellites acquiring original imagery. **(b)** Large-scale remote sensing images are sliced into smaller segments. **(c)** Annotated samples within the dataset, where the image portion visualizes object detection annotations, while the text portion serves as annotations for image classification. The last digit, highlighted in red, indicates the category.

Parameter	Specification
Number of Bands	32
Number of Spectral Channels	256
Spectral Range	400 nm–1000 nm
Spectral Resolution	~2.5 nm
True Color Combinations	Primary: B14, B7, B2 Alternative: B15, B5, B1

Table 2. Spectral characteristics.

satellites can be widely applied in fields such as land and resources, agriculture, forestry, animal husbandry, and fisheries, environmental protection, transportation, and smart cities. Specific applications include crop area statistics and yield estimation, environmental monitoring, disaster forecasting, and disaster assessment. The data collected by the satellites, upon reception and processing at ground stations, can generate high-precision hyperspectral images and spectral data. The standard processing workflow for Zhuhai-1 hyperspectral data includes steps such as radiometric calibration, atmospheric correction, and orthorectification, as illustrated in Fig. 3.

The Zhuhai-1 hyperspectral satellite data used in this study was obtained from the official data platform of Zhuhai Aerospace Microchips Science & Technology Co., Ltd. (<https://www.obtdata.com>). Researchers can access these data by registering on the platform and submitting data acquisition requests according to their research needs. The platform provides various levels of data products (L1A, L1B, etc.) with different processing levels. For academic research purposes, users may need to provide institutional information and research objectives during the application process. The authors obtained the L1B-level hyperspectral data used in this study through this official platform with appropriate research authorization. All data requests and processing follow the company’s data usage policies and licensing agreements.

During the process of decryption, decompression, and decoding, the primary focus is on the raw bitstream data demodulated by the ground demodulator. Based on the pre-agreed data format, auxiliary data and image data are parsed out to generate Level 0 data. Auxiliary data primarily include measurements from the satellite such as attitude, orbit, and time, which are utilized for subsequent geometric processing. Radiation processing is primarily used to correct inaccuracies in image grayscale response, ensuring it aligns as closely as possible with true grayscale values. However, due to the fact that radiation processing can, to a certain extent, compromise the spectral information of hyperspectral data, thereby affecting subsequent identification and analysis applications, hyperspectral radiation processing generally only involves high-precision systematic corrections, which are carried out based on high-accuracy radiometric calibration parameters. Geometric processing is primarily employed to correct inaccuracies in the positional grayscale of images, encompassing factors such as camera distortion, high-frequency vibrations, installation errors of devices, variations in exterior orientation elements, orbital and attitude errors, among others. It also involves utilizing auxiliary data to establish positioning models, including rigorous models and Rational Polynomial Coefficient (RPC) models. In the geometric processing of hyperspectral data, key aspects include geometric correction for single detectors to account for camera distortion and platform vibrations, mosaicing between multiple detectors, registration across all spectral bands, and

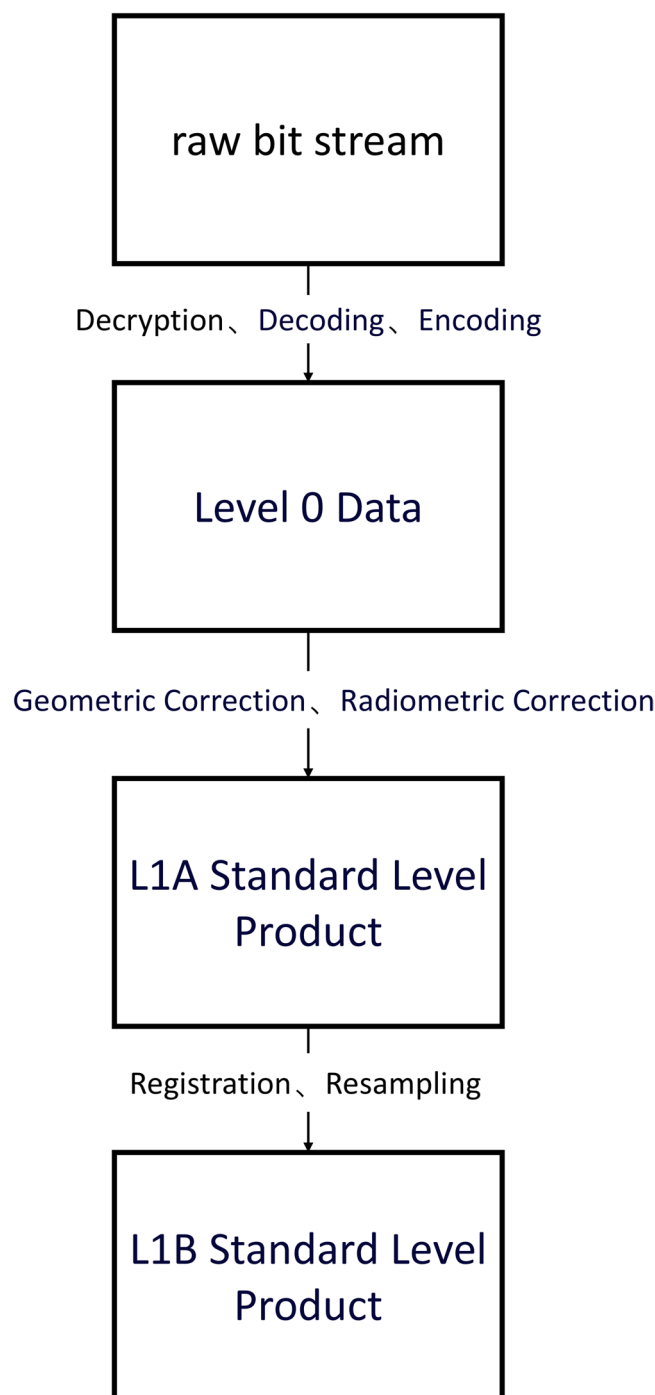


Fig. 3 Processing flowchart for hyperspectral standard products.

the construction of positioning models. The primary objectives of registration and resampling are to achieve a one-to-one correspondence of homologous points across the 32 spectral bands and to ensure that the geographic coverage of the imaging for all 32 bands is consistent. This process involves matching and cropping.

The classification of data products from the Zhuhai-1 hyperspectral remote sensing satellite is as follows:

- L0-level product (not released to users): The satellite transmits raw bitstream data to the ground station. After demodulation, descrambling, and decompression of the raw bitstream, L0-level product data is generated.
- L1A-level product (not released to users): This involves performing geometric correction and relative radiometric correction.
- L1B-level product: The L1B-level product undergoes full spectral band registration and provides 32-band image files, RPC files, metadata files, projection information files, and others.

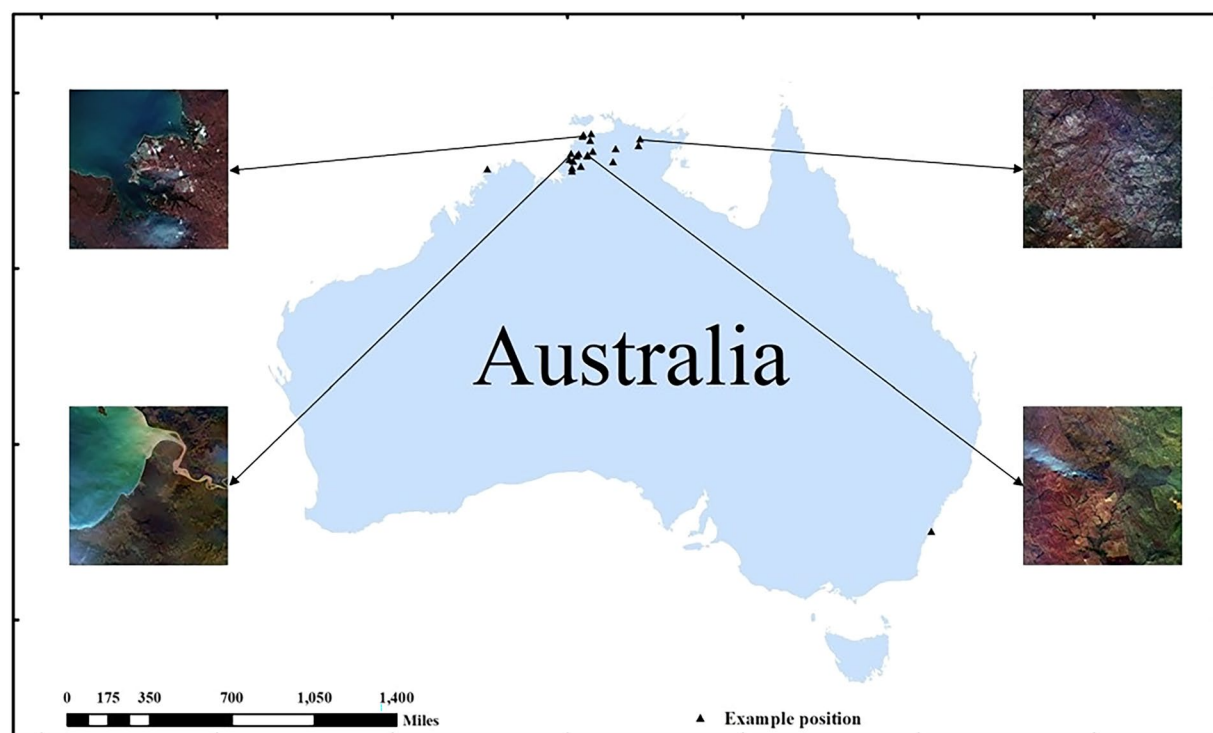


Fig. 4 Regional coverage of remote sensing imagery.

In our paper, we utilized the L1B-level product as our original imagery level.

Figure 3 displays the standard processing workflow for Zhuhai-1 hyperspectral data, which encompasses steps such as radiometric calibration, atmospheric correction, and orthorectification.

Study area. The dataset covers 22 locations in Australia, encompassing urban areas, mountainous regions, oceans, and other terrains, as illustrated in Fig. 4. Different terrain types, such as urban areas, mountains, and oceans, exhibit unique surface features and spectral characteristics. For instance, urban areas typically contain a variety of artificial structures and materials, resulting in complex and diverse spectral features. In contrast, mountainous regions may demonstrate varied spectral responses due to topographical variations and vegetation cover. This diversity increases the complexity of the dataset, providing a richer range of scenarios for model training and validation. However, it also elevates the difficulty of classification and detection tasks, ultimately enhancing the model's generalization capability.

Figure 4 showcases the dataset, which spans across 22 locations in Australia, encompassing diverse terrains including urban areas, mountainous regions, oceans, and other landscapes. The majority of the images are located in or around Darwin, in Australia's Northern Territory. These locations are primarily distributed around Darwin and extend to areas north and west of the city, forming a relatively concentrated yet spatially diverse set of samples. One image is situated in the arid region of northwestern Australia, an area with environmental conditions markedly different from the tropical, humid climate of Darwin. This region lies within the Western Desert of Australia and encompasses desert landscapes such as the Gibson Desert, Great Sandy Desert, and Little Sandy Desert. Another outlier corresponds to the Goulburn region in southeastern New South Wales. This area belongs to Australia's agricultural belt and is characterized by rolling hills and extensive farmland, representing a distinct land-use and ecological context compared to the northern and western sites.

The specific information for each remote sensing image is presented in Table 3. The Data column contains unique identifiers for each original remote sensing image, typically incorporating details such as the capture date, time, satellite ID, and sensor type. The naming convention is as follows: Satellite + ID + Receiving Station_ Receiving time_scene_level_sensor. Here, "Satellite" represents the type of remote sensing satellite within the Zhuhai-1 constellation, with "H" denoting the hyperspectral satellite. "ID" stands for the satellite number, specifically the hyperspectral satellite's ID. "Receiving Station" indicates the ground receiving station's number, while "Receiving time" specifies the time when the image was received on the ground. "Scene" refers to the imaging method of the hyperspectral satellite, where images are produced after full-track downlinking and cut into scenes of 50×50 km, with the scene number accumulated in the order of cutting. "Level" indicates the processing level of the image product, with L1B-level hyperspectral products currently provided to users. "Sensor" signifies that each hyperspectral satellite is equipped with three CMOS sensors, named CMOS1, CMOS2, and CMOS3, respectively. The Resolution column lists the spatial resolution for all images, with a ground sampling distance of 10 m for all, which facilitates clear identification of surface details and is particularly crucial for forest fire monitoring. The Fire Stage column describes the current status or phase of a fire event in terms of its activity level, development trend, and environmental impact. Specifically, the Active Burning Stage signifies that the fire is actively burning and spreading, while the Inactive Stage denotes that the fire has been fully extinguished with no signs of reignition.

Data	Fight time	Resolution	Fire stage
HEM1_20200623235326_0005_L1B_CMOS2	6/23/2020	10 m	Active Burning
HEM2_20200722222051_0003_L1B_CMOS2	7/22/2020	10 m	Active Burning
HEM2_20200722222051_0008_L1B_CMOS3	7/22/2020	10 m	Active burning
HEW1_20200622150241_0006_L1B_CMOS2	6/22/2020	10 m	Active burning
HFM2_20200827222226_0001_L1B_CMOS1	8/27/2020	10 m	Active burning
HFM2_20200827222226_0001_L1B_CMOS3	8/27/2020	10 m	Active burning
HFM2_20200827222226_0002_L1B_CMOS1	8/27/2020	10 m	Active burning
HFM2_20210516234906_0013_L1B_CMOS2	5/16/2021	10 m	Active burning
HFM2_20210516234906_0013_L1B_CMOS3	5/16/2021	10 m	Active burning
HFM2_20210516234906_0015_L1B_CMOS1	5/16/2021	10 m	Active burning
HFM2_20210516234906_0016_L1B_CMOS1	5/16/2021	10 m	Active burning
HFM2_20210516234906_0016_L1B_CMOS3	5/16/2021	10 m	Active burning
HGM1_20200718224680_0005_L1B_CMOS1	7/18/2020	10 m	Inactive Stage
HGM2_20200603235033_0005_L1B_CMOS2	6/03/2020	10 m	Inactive Stage
HGM2_20200603235033_0007_L1B_CMOS2	6/03/2020	10 m	Active burning
HGM2_20200621234228_0005_L1B_CMOS1	6/21/2020	10 m	Active burning
HGM2_20200621234228_0006_L1B_CMOS1	6/21/2020	10 m	Active burning
HGZ3_20200824012507_0005_L1B_CMOS3	8/24/2020	10 m	Active burning
HGZ3_20200824012507_0006_L1B_CMOS3	8/24/2020	10 m	Active burning
HHM1_20210819234751_0003_L1B_CMOS2	8/19/2021	10 m	Active burning
HHW1_20201002162230_0006_L1B_CMOS2	10/02/2020	10 m	Active burning

Table 3. Specific information of remote sensing imagery in OHID-FF dataset.

Data tilling. Large-scale remote sensing images typically encompass vast amounts of data. Directly processing them in their entirety not only consumes considerable time and effort but also places enormous demands on computer hardware resources. It is also worth noting that most of the original images in the dataset suffer from a problem of extremely small fire regions. Take the image HFM2_20200827222226_0002_L1B_CMOS1 as an example, where the proportion of fire pixels accounts for only about 0.2%. As a result, after slicing processing, there will be a severe imbalance in the ratio of positive and negative samples.

To enhance computational efficiency, ensure compliance with neural network input size requirements, maintain the coherence of image information, and further optimize memory usage, we adopted slicing processing measures for Zhuhai-1 imagery. By implementing the slicing strategy, we can scientifically divide these large-scale images into multiple small-sized image segments. This approach effectively simplifies the processing workflow and significantly reduces the demand for hardware resources. Additionally, slicing processing facilitates the management and storage of image data, laying a solid foundation for subsequent data analysis and application work. The specific slicing operation is illustrated in Fig. 5. We conducted slice preprocessing on the acquired original satellite images of fire, resizing all original images to 512*512 pixels with an overlap of 256 pixels. For fire regions, we employed resampling methods to increase the number of samples, while for non-fire regions, we implemented undersampling strategies to reduce redundant background samples. Through these processing techniques, we constructed a dataset with a balanced ratio of positive and negative samples.

Figure 5 demonstrates a schematic diagram of large-scale remote sensing imagery being divided into overlapping windows.

Data labeling and analysis. As shown in Fig. 6, the annotations for forest fire points adhere to the YOLO²⁸ format, encompassing only one category: forest fire points. Furthermore, the last digit in the filename of each image serves as an indicator of whether the image contains fire points, with “1” denoting the presence of fire points and “0” indicating their absence. Consequently, this dataset is not only suitable for forest fire detection tasks but can also be utilized for research in image classification.

After completing the dataset preparation, we conducted a detailed statistical analysis on it. We classified and counted the object bounding boxes in the dataset into three categories: large, medium, and small. According to the standards of the COCO dataset²⁹, small objects (S) refer to those whose bounding box pixel area accounts for less than 0.5% of the total image area, medium objects (M) range between 0.5% and 1%, and the rest are considered large objects (L). The statistical results revealed that the number of large objects in the dataset is significantly higher than that of medium and small objects, posing a challenge for the object detection task of this dataset, as detecting small objects is typically more difficult.

As illustrated in Fig. 7, the dataset maintains a basic balance between positive and negative samples, which is conducive to enhancing the model’s performance and robustness in classification tasks.

Figure 7 presents a pie chart that statistically summarizes the proportion of forest fire targets and background targets within the dataset. The orange section represents the background, while the blue section represents the forest fire targets. The numerical values indicate the specific proportions of each category.

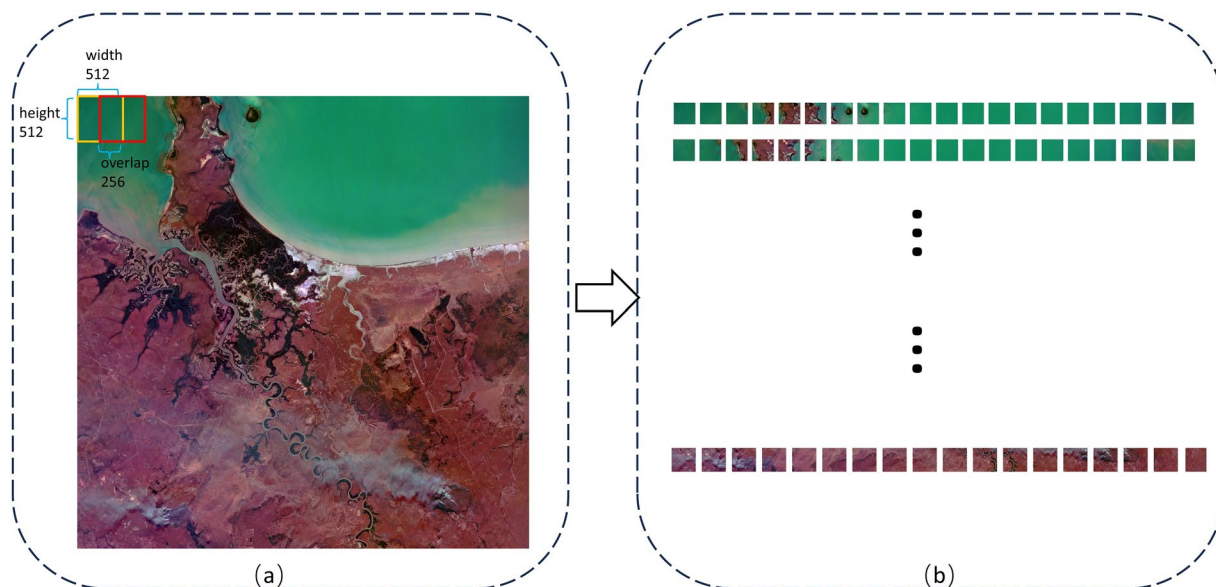


Fig. 5 Tiling of remote sensing imagery.

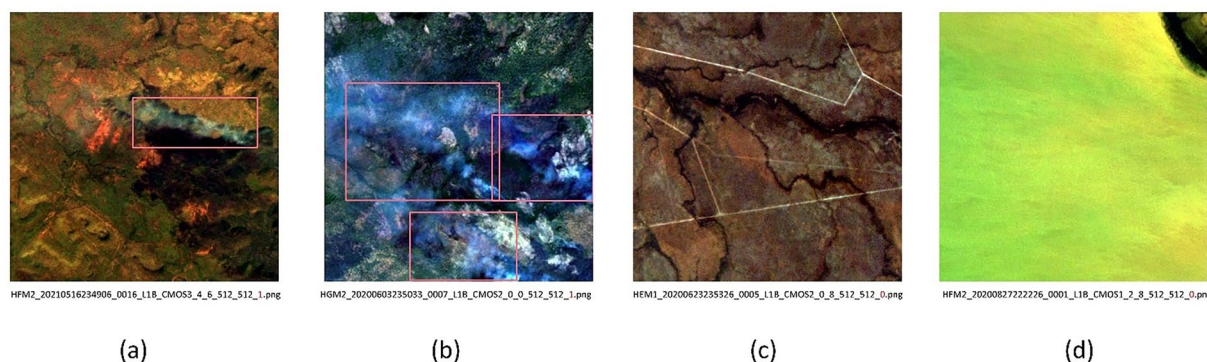


Fig. 6 Visualization of image annotation: (a,b) Samples of images containing fire targets. (c,d) Samples of background images.

Data Records

The OHID-FF dataset²⁰ is available at Figshare repository <https://doi.org/10.6084/m9.figshare.28218485.v6>. It is distributed under the CC BY 4.0 license, allowing unrestricted reuse. The organizational structure of the folders is illustrated at the Fig. 8.

Figure 8 showcases the specific file structure in which the dataset has been stored. The image files at the OHID-FF dataset were initially collected from Zhuhai-1 hyperspectral satellite constellation. These original data images, each measuring 5056×5056 pixels, total 22 in number and are stored in the “tif” folder. The structure of the “YOLODataset” folder comprises the following:

1. “images” subfolder containing remotely sensed images that have undergone slicing processing.
2. “labels” subfolder storing the corresponding label files for these images, named according to a uniform format (e.g., HEM1_20200623235326_0005_L1B_CMOS2_0_8_512_512_0.*). Here, “HEM1_20200623235326_0005_L1B_CMOS2” indicates the source of the satellite image, “0_8_512_512” specifies the position of the slice within the original image, and the final digit signifies the presence of a specific category. All labels adopt the “xywh” format.
3. “viz” subfolder containing visualizations of the dataset labels.
4. “classes.txt” file listing the category names present in the dataset.
5. “dataset.yaml” file providing specific paths to the entire dataset, although these paths may require adjustment for use.

This organization ensures efficient access and utilization of the dataset for various research purposes.

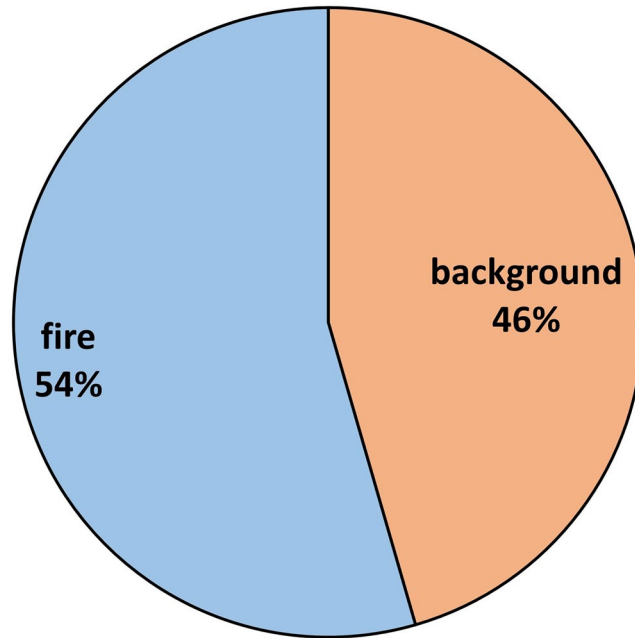


Fig. 7 Proportion of each category in the OHID-FF dataset.

Technical Validation

Algorithms adopted. The study leveraged a combination of state-of-the-art deep learning algorithms tailored for hyperspectral image classification and forest fire detection tasks.

To evaluate the effectiveness and generalization of the proposed dataset for fire detection in remote sensing images, a range of representative deep learning models were selected. These include ResNet18¹⁴ and ResNet50¹⁴, which address gradient vanishing through residual connections, and VGG16³⁰, known for its depth and small convolutional kernels. Logistic regression³¹ was included as a baseline for comparison, while lightweight models like MobileNetV2³² and ShuffleNetV2³³ were chosen for their efficiency in real-time applications. Additionally, LeNet5³⁴ represented early CNN successes, and Flame_one_stream²⁶ was specifically included for flame recognition. DenseNet121³⁵ and InceptionV3³⁶ were also tested for their feature reuse and multi-scale processing capabilities.

Evaluation metrics. The performance of models was evaluated using standardized metrics for classification tasks: Accuracy, F1-score, Precision, and Recall.

Accuracy: the proportion of correctly classified instances (both true positives and true negatives) out of the total samples:

$$\text{Accuracy} = \sum_{i=1}^N \frac{TP_i + TN_i}{TP_i + TN_i + FP_i + FN_i} \quad (1)$$

F1-Score: the harmonic mean of Precision and Recall, balancing both metrics. It is especially useful for imbalanced datasets:

$$\text{Macro-F1} = \frac{1}{N} \sum_{i=1}^N 2 \times \frac{\text{Precision}_i \times \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i} \quad (2)$$

$$\text{Micro-F1} = 2 \times \frac{\sum_{i=1}^N TP_i}{\sum_{i=1}^N (TP_i + FP_i) + \sum_{i=1}^N (TP_i + FN_i)} \quad (3)$$

Precision: the ratio of correctly predicted positive instances (TP) to all predicted positive instances (TP + FP). It measures the model's ability to avoid false positives:

$$\text{Macro-Precision} = \frac{1}{N} \sum_{i=1}^N \frac{TP_i}{TP_i + FP_i} \quad (4)$$

$$\text{Micro-Precision} = \frac{\sum_{i=1}^N TP_i}{\sum_{i=1}^N (TP_i + FP_i)} \quad (5)$$



Fig. 8 Organizational structure of the folders at OHID-FF dataset.

Recall: The ratio of correctly predicted positive instances (TP) to all actual positive instances (TP + FN). It quantifies the model's ability to identify all relevant cases

$$\text{Macro-Recall} = \frac{1}{N} \sum_{i=1}^N \frac{TP_i}{TP_i + FN_i} \quad (6)$$

$$\text{Micro-Recall} = \frac{\sum_{i=1}^N TP_i}{\sum_{i=1}^N (TP_i + FN_i)} \quad (7)$$

Symbol Definitions:

- TP_i : True Positives for class i ;
- FP_i : False Positives for class i ;
- FN_i : False Negatives for class i ;
- N : Total number of classes;
- *Macro*: Metrics are calculated by averaging over all classes (equal weight per class);
- *Micro*: Metrics are calculated globally by aggregating TP , FP , and FN across all classes.

Experimental settings and results. We evaluate a suite of deep neural networks on the OHID-FF dataset for the task of binary fire/non-fire image classification. The dataset comprises 1,197 image patches, each of size 512×512 pixels. The class distribution is imbalanced, with 647 fire samples (54.0%) and 550 non-fire samples (46.0%) - consistent with the statistics illustrated in Fig. 7.

To ensure a fair and reproducible evaluation, the dataset was partitioned into a training set (597 images) and an independent test set (600 images) using stratified random sampling based on the fire/non-fire label. This strategy preserves the original class proportions in both subsets and prevents data leakage. No cross-validation was performed. The test set remains completely held-out until final evaluation.

Model	Accuracy	F1	Precision	Recall	Average Method
Resnet18 ¹⁴	0.918	0.918	0.918	0.924	macro
		0.918	0.919	0.918	micro
VGG16 ³⁰	0.544	0.352	0.273	0.5	macro
		0.544	0.545	0.544	micro
Logistic regression ³¹	0.689	0.677	0.686	0.675	macro
		0.689	0.689	0.689	micro
MobileNetV2 ³²	0.985	0.985	0.984	0.986	macro
		0.985	0.985	0.985	micro
LeNet5 ³⁴	0.529	0.346	0.265	0.5	macro
		0.529	0.529	0.529	micro
Flame_one_Stream ²⁶	0.880	0.878	0.878	0.879	macro
		0.880	0.880	0.880	micro
Densenet121 ³⁵	0.973	0.973	0.972	0.974	macro
		0.973	0.973	0.973	micro
Shufflenetv2 ³³	0.967	0.966	0.966	0.967	macro
		0.967	0.967	0.967	micro
Inceptionv3 ³⁶	0.990	0.990	0.989	0.990	macro
		0.990	0.990	0.990	micro
Resnet50 ¹⁴	0.970	0.970	0.969	0.972	macro
		0.970	0.970	0.970	micro

Table 4. Performance comparison of different models on OHID-FF dataset, RGB mode.

All models were adapted to the binary fire/non-fire classification task by replacing the original classification head with a two-class output layer, while keeping the backbone architecture and three-channel RGB input unchanged. The final layer was followed by a softmax activation to produce class probabilities.

Where available, backbones were initialized from official ImageNet pre-trained checkpoints. To account for stochasticity in training, each model was trained independently three times on the same train/test split, and the final performance metrics represent the arithmetic mean across these runs. We conducted benchmark experiments on the OHID-FF dataset using multiple deep neural networks. The performance metrics results are shown in Table 4.

Analysis of experimental results. As shown in Table 4, models such as Inceptionv3 achieved an F1 score of 99%, demonstrating the dataset's suitability for training robust hyperspectral classification systems. Lightweight architectures like MobileNetV2 also performed exceptionally well (F1 = 98.5%), highlighting the potential for deploying efficient real-world monitoring solutions. As shown in Fig. 9, which displays the normalized confusion matrices of all benchmark models on the OHID-FF test set, the rows represent the ground-truth classes and the columns the predicted classes, with the two categories being fire and non-fire. These confusion matrices also indicate that Inceptionv3 and MobileNetV2 have the best performance. In contrast, traditional methods like logistic regression struggled significantly (F1 = 67.7%), underscoring the necessity of deep learning for hyperspectral fire analysis.

Notably, models with insufficient spectral complexity (e.g., VGG16 and LeNet5) exhibited severe underfitting (precision < 30%). From Fig. 9, we can find that these two models' confusion matrices show that they can't distinguish these two labels, likely due to their inability to process the dataset's 32-band spectral features. This observation emphasizes OHID-FF's fundamental advantage: it uniquely challenges models to leverage high-dimensional spectral data rather than simplistic RGB approximations. Besides, this dataset's coverage of urban, mountainous, and oceanic terrains improved model generalization. However, there're limitations and challenges: models struggled with small fire points, highlighting the need for specialized architectures or attention mechanisms; data limited to Australian regions may reduce applicability to other ecosystems; high-resolution hyperspectral processing required significant GPU resources.

Usage Note

OHID-FF dataset²⁰ described here is available from <https://doi.org/10.6084/m9.figshare.28218485.v6>. It offers flexibility for researchers aiming to study HSI classification and forest fire detection.

The OHID-FF dataset²⁰ comprises 22 original wildfire remote sensing scenes (individual scene dimensions: 5056 × 5056 pixels, 32 spectral bands) captured across 22 distinct Australian locations. To adapt for fire detection tasks, the original imagery was segmented into 512 × 512 pixel sub-images with fire points annotated using the YOLO algorithm. Data acquisition by the Zhuhai-1 satellites employed a Ground Sampling Distance (GSD) of 10 meters per pixel, achieving optimal remote sensing capabilities while preserving spatial detail.

The OHID-FF dataset²⁰ specifically tailored for forest fire detection and classification, is a treasure trove of data with numerous advantages. As a high-quality hyperspectral image dataset, it offers rich data volume, high spectral and spatial resolution, and diverse scene coverage. It covers different terrain types in Australia, such as urban areas, mountains, and oceans, exhibit unique surface features and spectral characteristics. This diversity

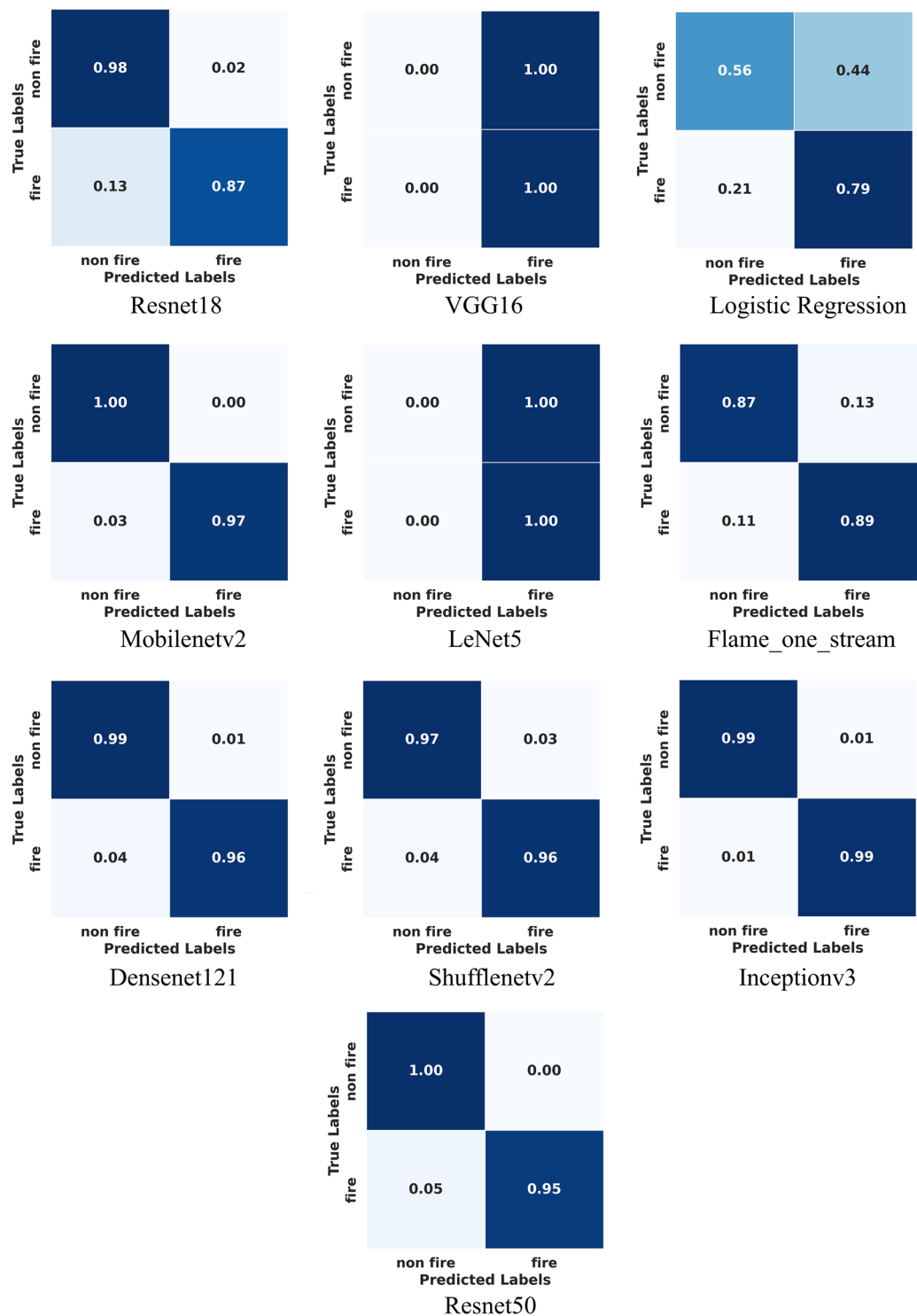


Fig. 9 Normalized Confusion Matrices of different models on OHID-FF dataset.

increases the complexity of the dataset, providing a richer range of scenarios for model training and validation. It also elevates the difficulty of classification and detection tasks, ultimately enhancing the model's generalization capability. These attributes make it an invaluable resource not only for forest fire-related research but also for extending its application to other fields such as environmental monitoring, detecting of nature hazards, disaster assessment, resource management, zone planning providing richer data support for relevant research endeavors. By providing robust support for deep learning model training, it has the potential to significantly advance the state-of-the-art in these areas. Furthermore, researchers can leverage this dataset to explore more advanced deep learning models and algorithms, aiming to enhance the classification and detection accuracy of the OHID-FF dataset²⁰, especially when dealing with small area analysis and imbalanced datasets.

However, it is important to acknowledge that the dataset is not without its limitations. The main drawback of the dataset is that it is only focused on the task of the binary classification of fire/non-fire areas, and therefore of limited interest for other uses, such as detecting land cover types, change detection, etc. Some of the challenges associated with this dataset include data imbalance, which can pose difficulties for training accurate models; high computational resource requirements, which may limit its accessibility to researchers with limited hardware; limited annotation diversity, which could affect the generalization capability of trained models; and restricted geographical coverage, which may not fully represent the diverse conditions found in different forest ecosystems. These limitations need to be carefully considered and weighed against the dataset's advantages when planning research projects. Despite these challenges, the overall strengths of the dataset make it a pivotal resource in forest fire monitoring and related fields, with considerable potential for fostering innovative research and applications.

For future study, we will expand the capacity of the OHID-FF²⁰ from 2 GB level to 10 GB level. Additionally, we will provide more granular annotations for forest fire scenes (such as distinguishing between smoke and ignition points) to address the complexity and confusion in such areas. We will also provide data for a wider range of regions to cover more diverse landforms and landscapes. Furthermore, we will develop more efficient and accurate classification and detection algorithms as benchmark methods based on the characteristics of the OHID-FF²⁰, in order to meet increasingly diverse application requirements. To enable more accurate predictions of fire spread trends and fine-grained analysis of complex scenarios, pixel-level annotations will be provided in subsequent updates.

Data availability

The OHID-FF dataset is available at Figshare repository <https://doi.org/10.6084/m9.figshare.28218485.v6>.

Code availability

The codes for dataset partitioning and train/test setup is available in our GitHub repository: <https://github.com/hrnavy/OHID-FF/tree/main>. Additional scripts for models updating, training, evaluation are also provided to support end-to-end experimentation.

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References

- Chen, X., Liu, L., Li, J., Ow, W. & Zhang, Y. Application and research progress of fire monitoring using satellite remote sensing. *National Remote Sensing Bulletin*. **24**(5), 531–542 (2020).
- Chen, Y., Ma, X., Xu, Y., Wang, Y. & Wang, Y. Methods for the application of topography and NDVI in re-identification of remote sensing-based monitoring of forest fires. *Remote Sensing for Land & Resources*. **34**(3), 88–96 (2022).
- Qin, X., Li, X., Liu, S. & Lu, Q. Forest fire early warning and monitoring techniques using satellite remote sensing in China. *National Remote Sensing Bulletin*. **24**(5), 511–520 (2020).
- Seydi, S. T., Saeidi, V., Kalantar, B., Halin, A. A. & Ueda, N. Fire-Net: A Deep Learning Framework for Active Forest Fire Detection. *Journal of Sensors*. **2022**, 1–14 (2022).
- San-Miguel-Ayanz, J. & Ravail, N. Active fire detection for fire emergency management: Potential and limitations for the operational use of remote sensing. *Natural Hazards*. **35**(3), 361–376 (2005).
- Liu, S., Li, X., Qin, X., Sun, G. & Liu, Q. Adaptive threshold method for active fire identification based on GF-4 PMI data. *National Remote Sensing Bulletin*. **24**(3), 215–225 (2020).
- Zhang, N., Sun, L. & Sun, Z. GF-4 Satellite Fire Detection with an improved Contextual algorithm. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*. **15**, 163–172 (2021).
- Flasse, S. P. & Ceccato, P. A. contextual algorithm for AVHRR fire detection. *International Journal of Remote Sensing*. **17**(2), 419–424 (1996).
- Preeti, T., Kanakaraddi, S., Beelagi, A., Malagi, S. & Sudi, A. Forest fire prediction using machine learning techniques. *2021 International Conference on Intelligent Technologies (CONIT)*. IEEE, 1–6 (2021).
- Lu, D. & Weng, Q. A survey of image classification methods and techniques for improving classification performance. *International journal of Remote sensing*. **28**(5), 823–870 (2007).
- Luo, H. L. & Chen, H. K. Survey of object detection based on deep learning. *Acta Electronica Sinica*. **48**(6), 1230–1239 (2020).
- Lan, X., Wu, S., Fu, B. & Qin, X. Survey on Deep Learning in Oriented Object Detection in Remote Sensing Images. *Journal of Frontiers of Computer Science and Technology*. **18**(4), 861–877 (2024).
- Zou, Z., Chen, K., Shi, Z. & Ye, J. Object detection in 20 years: A survey. *Proceedings of the IEEE*. **111**(3), 257–276 (2023).
- He, K., Zhang, X., Ren, S. & Sun, J. Deep residual learning for image recognition. *Proceedings of the IEEE conference on computer vision and pattern recognition*. 770–778 (2016).
- Dosovitskiy, A. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929* (2020).
- Zhang, X. *et al.* Remote sensing object detection meets deep learning: A metareview of challenges and advances. *IEEE Geoscience and Remote Sensing Magazine*. **99**, 2–38 (2023).
- Ren, S., He, K., Girshick, R., Sun, J. & Faster, R.-C. N. N. Towards real-time object detection with region proposal networks. *IEEE transactions on pattern analysis and machine intelligence*. **39**(6), 1137–1149 (2016).
- Sapkota, R. *et al.* YOLOv10 to Its Genesis: A Decadal and Comprehensive Review of The You Only Look Once Series. *arXiv preprint arXiv:2406.19407* (2024).
- Jiang, P., Ergu, D., Liu, F., Ying, C. & Ma, B. A Review of Yolo algorithm developments. *Procedia computer science*. **199**(11), 1066–1073 (2022).
- Chen, X. & Deng, J. OHID-FF. *figshare* <https://doi.org/10.6084/m9.figshare.28218485.v6> (2025).
- Wang, M. *et al.* An open flame and smoke detection dataset for deep learning in remote sensing based fire detection. *Geo-spatial Information Science*, 1–16 (2024).
- Hu, X. *et al.* An optimized smoke segmentation method for forest and grassland fire based on the UNet framework. *Fire*. **7**(3):68, 1–18 (2024).
- Wang, Z. *et al.* Semantic segmentation and analysis on sensitive parameters of forest fire smoke using smoke-unet and landsat-8 imagery. *Remote Sensing*. **14**(1), 1–19 (2021). 45.

24. El-Madafri, I., Peña, M. & Olmedo-Torre, N. The Wildfire Dataset: Enhancing Deep Learning-Based Forest Fire Detection with a Diverse Evolving Open-Source Dataset Focused on Data Representativeness and a Novel Multi-Task Learning Approach. *Forests*. **14**(9):1697, 1–25 (2023).
25. Kyrkou, C. & Theocharides, T. Deep-Learning-Based Aerial Image Classification for Emergency Response Applications Using Unmanned Aerial Vehicles. *CVPR workshops*. 517–525 (2019).
26. Shamsoshoara, A. *et al.* Aerial imagery pile burn detection using deep learning: The FLAME dataset. *Computer Networks*. **193**(4), 108001 (2021).
27. Chen, X. *et al.* Wildland fire detection and monitoring using a drone-collected rgb/ir image dataset. *IEEE Access*. **10**, 121301–121317 (2022).
28. Redmon, J., Divvala, S., Girshick, R. & Farhadi, A. You only look once: Unified, real-time object detection. *Proceedings of the IEEE conference on computer vision and pattern recognition*. (2016).
29. Lin, T.-Y. *et al.* Microsoft coco: Common objects in context. *Proceeding of the Computer Vision–ECCV 2014: 13th European Conference*. Springer International Publishing, 740–755 (2014).
30. Simonyan, K. & Zisserman, A. Very Deep Convolutional Networks for Large-Scale Image Recognition. *arXiv:1409.1556* (2014).
31. Pal, M. Multinomial logistic regression-based feature selection for hyperspectral data. *International Journal of Applied Earth Observation and Geoinformation*. **14**, 214–220 (2012).
32. Sandler, M. *et al.* MobileNetV2: Inverted Residuals and Linear Bottlenecks. *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 4510–4520 (2018).
33. Ma, N., Zhang, X., Zheng, X.-T. & Sun, J. ShuffleNet V2: Practical Guidelines for Efficient CNN Architecture Design. *Computer Vision – ECCV 2018*. 122–138. (2018).
34. Lecun, Y., Bottou, L., Bengio, Y. & Haffner, P. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*. 2278–2324 (1996).
35. Gao, H., Zhuang Liu, Z. & Weinberger, K. Q. Densely Connected Convolutional Networks. *2017 IEEE Conference on Computer Vision and Pattern Recognition*. 2261–2269 (2016).
36. Szegedy, C., Vanhoucke, V., Ioffe, S., Shlens, J. & Wojna, Z. Rethinking the Inception Architecture for Computer Vision. *2016 IEEE Conference on Computer Vision and Pattern Recognition*. 2818–2826 (2016).

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Author contributions

The original data with quality and data standard control, data labeling and presentation was provided by X.C., Z.Y., J.W., J.D. and X.J. A.M., S.G., J.Y., Y.S. and Y.C. contributed to conceptualization, methodology and formal analysis. A.M., X.C. and A.D. have performed experimental studies and analysis experiments. A.M. and X.C. wrote the manuscript. A.M., X.C., S.G. and Y.C. contributed to revision of the manuscript to its final version.

Competing interests

The authors declare no competing interests.

Additional information

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